Genesis_Medicine - 43_r17_chromanol_generative_atlas

R17 Constrained Generative Chromanol Analog Atlas: Photosafety-Gated Boltz-2 Cofolding and 10-60 ns MD Robustness

Abstract

R17 extends the R16 topical chromanol program with a constrained generative analog atlas. The goal was not an unconstrained novelty claim, but a disciplined hit-to-lead expansion that keeps the chromanol core compact while varying substitution patterns and tracking target, photosafety, and prior-art gates. The atlas contains `240` Boltz-2 cofold rows across balanced skin-relevant targets. Photosafety-green candidates reached affinity probabilities up to `0.672`, while aryl-halogen candidates were retained only with explicit photosafety review labels. The MD ladder covers top, next, and expanded green-target panels through 10, 30, and 60 ns. Expanded 60 ns status is `3/3` ok, so this manuscript status is `complete`. All findings are in silico only and remain Markush/FTO and wet-lab validation pending.

Keywords: chromanol, generative design, Boltz-2, OpenMM, photosafety, prior-art gate, in silico

1. Research Question

Can a constrained R17 chromanol analog generator produce target-relevant, photosafety-gated candidates that remain stable through staged MD robustness checks without overclaiming novelty, freedom to operate, or clinical efficacy?

2. Data Sources

file	role
`pilot/cpu_meaningful/r17_chromanol_generative_batch*_cofold.csv`	240-row Boltz-2 cofold atlas
`pilot/md_r17_chromanol_generative_top_green_*/summary.json`	top green-target 10/30/60 ns MD ladder
`pilot/md_r17_chromanol_generative_next_green_*/summary.json`	next green-target 10/30/60 ns MD ladder
`pilot/md_r17_chromanol_generative_expanded_green_*/summary.json`	expanded green-target 10/30/60 ns MD ladder
`pilot/cpu_meaningful/precompute_prior_art_gate.csv`	technical prior-art and Markush pre-gate

3. Results

3.1 Cofold atlas by target

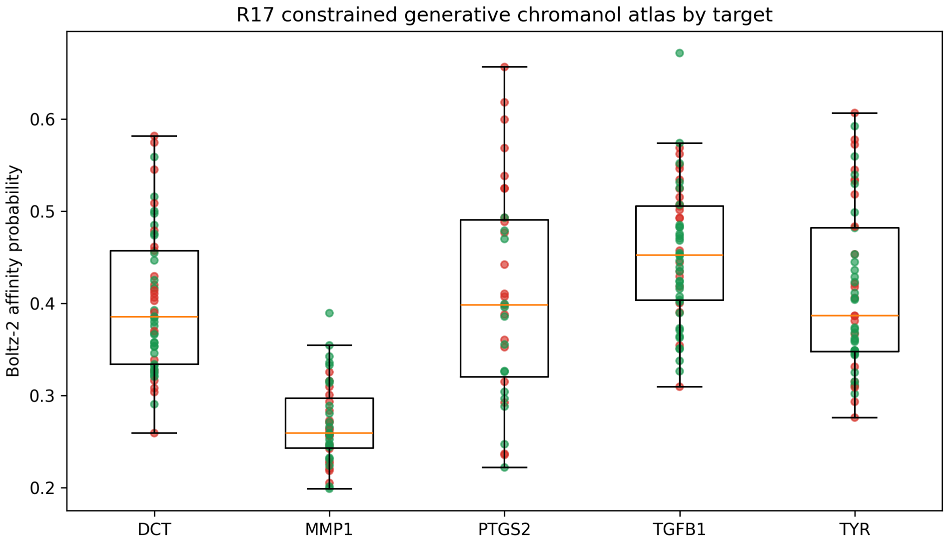

Figure 1. Distribution of R17 cofold affinity probabilities by target. Green points indicate `photosafety_proxy=none_detected`; red points indicate aryl-halogen review.

3.2 Top photosafety-green candidates

job_id	target	design	affinity probability	cLogP	QED	photosafety proxy
r17_15_tgfb1_chromanol_arom9_oh_tgfb1	TGFB1	chromanol_arom9_OH_tgfb1	0.672	0.64	0.576	none_detected
r17_30_tyr_chromanol_arom6_arom9_f_f_tyr	TYR	chromanol_arom6+arom9_F+F_tyr	0.592	1.21	0.741	none_detected
r17_16_tgfb1_chromanol_arom10_oh_tgfb1	TGFB1	chromanol_arom10_OH_tgfb1	0.574	0.64	0.616	none_detected
r17_22_tyr_chromanol_arom6_f_tyr	TYR	chromanol_arom6_F_tyr	0.559	1.07	0.709	none_detected
r17_29_dct_chromanol_arom6_arom9_f_f_dct	DCT	chromanol_arom6+arom9_F+F_dct	0.559	1.21	0.741	none_detected
r17_20_tgfb1_chromanol_arom9_f_tgfb1	TGFB1	chromanol_arom9_F_tgfb1	0.552	1.07	0.709	none_detected
r17_16_tyr_chromanol_arom9_f_tyr	TYR	chromanol_arom9_F_tyr	0.540	1.07	0.709	none_detected
r17_16_tgfb1_chromanol_arom6_arom9_f_ome_tgfb1	TGFB1	chromanol_arom6+arom9_F+OMe_tgfb1	0.532	1.08	0.795	none_detected
r17_08_tyr_chromanol_arom6_arom10_f_me_tyr	TYR	chromanol_arom6+arom10_F+Me_tyr	0.530	1.38	0.739	none_detected
r17_10_tgfb1_chromanol_arom9_me_tgfb1	TGFB1	chromanol_arom9_Me_tgfb1	0.525	1.24	0.707	none_detected

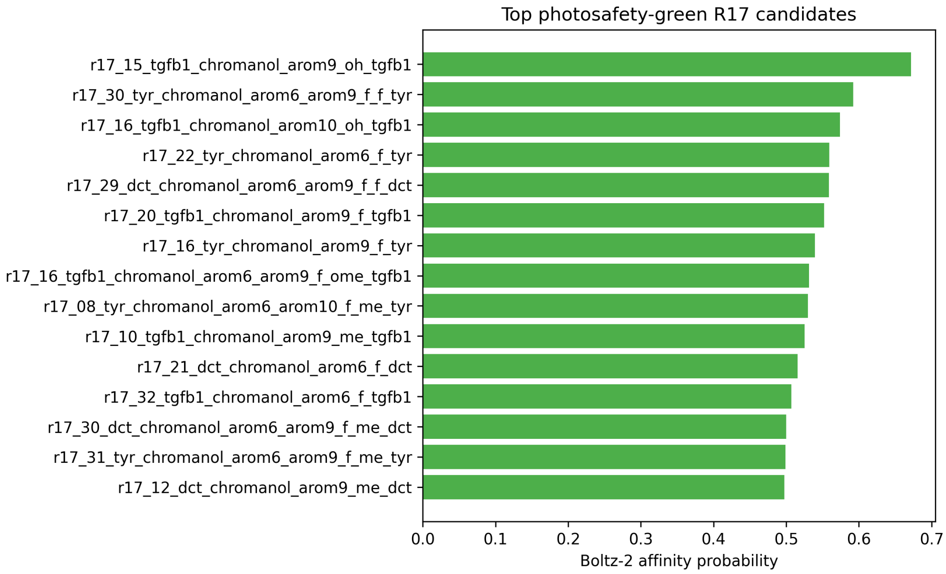

Figure 2. Top photosafety-green R17 candidates by Boltz-2 affinity probability.

3.3 Staged MD robustness

The R17 staged MD ladder currently contains 27 ok rows. The completed top and next green-target 60 ns panels were stable. The expanded 60 ns panel is the final promotion gate and is 3/3 complete in this manuscript snapshot.

panel	target	analog	affinity probability	mean RMSD A	last-third RMSD A	max RMSD A
expanded_10ns	DCT	chromanol_arom6+arom9_F+F_dct	0.559	0.49	0.59	1.00
expanded_10ns	TYR	chromanol_arom6+arom9_F+F_tyr	0.592	0.69	0.25	1.26
expanded_10ns	TYR	chromanol_arom6_F_tyr	0.559	0.51	0.52	0.70
expanded_30ns	DCT	chromanol_arom6+arom9_F+F_dct	0.559	0.33	0.21	0.83
expanded_30ns	TYR	chromanol_arom6+arom9_F+F_tyr	0.592	0.28	0.37	0.73
expanded_30ns	TYR	chromanol_arom6_F_tyr	0.559	0.30	0.23	0.61
expanded_60ns	DCT	chromanol_arom6+arom9_F+F_dct	0.559	0.51	0.48	0.76

panel	target	analog	affinity probability	mean RMSD A	last-third RMSD A	max RMSD A
expanded_60ns	TYR	chromanol_arom6+arom9_F+F_tyr	0.592	0.42	0.30	1.05
expanded_60ns	TYR	chromanol_arom6_F_tyr	0.559	0.48	0.46	0.72
next_10ns	DCT	chromanol_arom9+arom10_Cl+Cl_dct	0.575	0.51	0.60	0.87
next_10ns	DCT	chromanol_arom6+arom9_F+Me_dct	0.500	0.51	0.53	0.80
next_10ns	TYR	chromanol_arom9+arom10_F+Cl_tyr	0.578	0.51	0.56	0.76
next_30ns	DCT	chromanol_arom9+arom10_Cl+Cl_dct	0.575	0.53	0.54	0.70
next_30ns	DCT	chromanol_arom6+arom9_F+Me_dct	0.500	0.50	0.34	0.86
next_30ns	TYR	chromanol_arom9+arom10_F+Cl_tyr	0.578	0.51	0.51	0.68
next_60ns	DCT	chromanol_arom9+arom10_Cl+Cl_dct	0.575	0.53	0.70	1.10
next_60ns	DCT	chromanol_arom6+arom9_F+Me_dct	0.500	0.49	0.52	1.12
next_60ns	TYR	chromanol_arom9+arom10_F+Cl_tyr	0.578	0.97	0.97	1.23
top_10ns	DCT	chromanol_arom6+arom9_Cl+Cl_dct	0.582	0.71	0.69	0.94
top_10ns	TYR	chromanol_arom6+arom9_F+Cl_tyr	0.607	0.50	0.49	1.05
top_10ns	TYR		0.540	0.50	0.74	1.01

panel	target	analog	affinity probability	mean RMSD A	last-third RMSD A	max RMSD A
		chromanol_arom9_F_tyr				
top_30ns	DCT	chromanol_arom6+arom9_Cl+Cl_dct	0.582	0.50	0.73	0.98
top_30ns	TYR	chromanol_arom6+arom9_F+Cl_tyr	0.607	0.31	0.20	0.87
top_30ns	TYR	chromanol_arom9_F_tyr	0.540	0.53	0.55	0.70
top_60ns	DCT	chromanol_arom6+arom9_Cl+Cl_dct	0.582	0.54	0.55	0.72
top_60ns	TYR	chromanol_arom6+arom9_F+Cl_tyr	0.607	0.50	0.56	0.94
top_60ns	TYR	chromanol_arom9_F_tyr	0.540	0.55	0.62	0.82

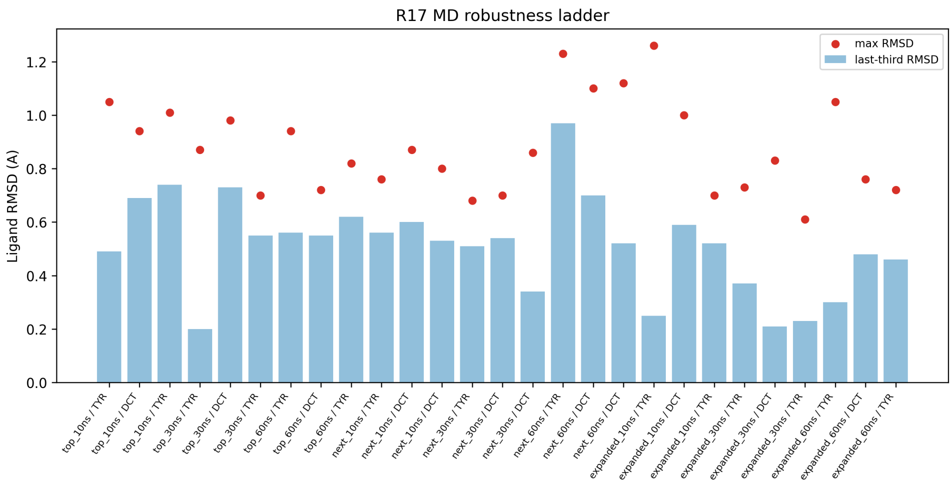

Figure 3. R17 MD robustness ladder across 10, 30, and 60 ns panels.

3.4 Expanded 60 ns final gate

name	target	analog	mean RMSD A	last-third RMSD A	max RMSD A
r17_30_tyr_chromanol_arom6_arom9_f_f_tyr__chromanol_arom6_arom9_F_F_tyr__60ns	TYR	chromanol_arom6+arom9_F+F_tyr	0.42	0.30	1.05

name	target	analog	mean RMSD A	last-third RMSD A	max RMSD A
r17_29_dct_chromanol_arom6_arom9_f_f_dct__chromanol_arom6_arom9_F_F_dct__60ns	DCT	chromanol_arom6+arom9_F+F_dct	0.51	0.48	0.76
r17_22_tyr_chromanol_arom6_f_tyr__chromanol_arom6_F_tyr__60ns	TYR	chromanol_arom6_F_tyr	0.48	0.46	0.72

3.5 Prior-art and Markush discipline

The R17 prior-art precompute gate counts are `{'hold_expensive_compute_until_markush_review': 240}`. This gate does not kill the scientific atlas, but it blocks stronger commercial language. A PubChem no-hit result is not freedom to operate, because Markush and use claims can still cover unmade or unlisted analogs.

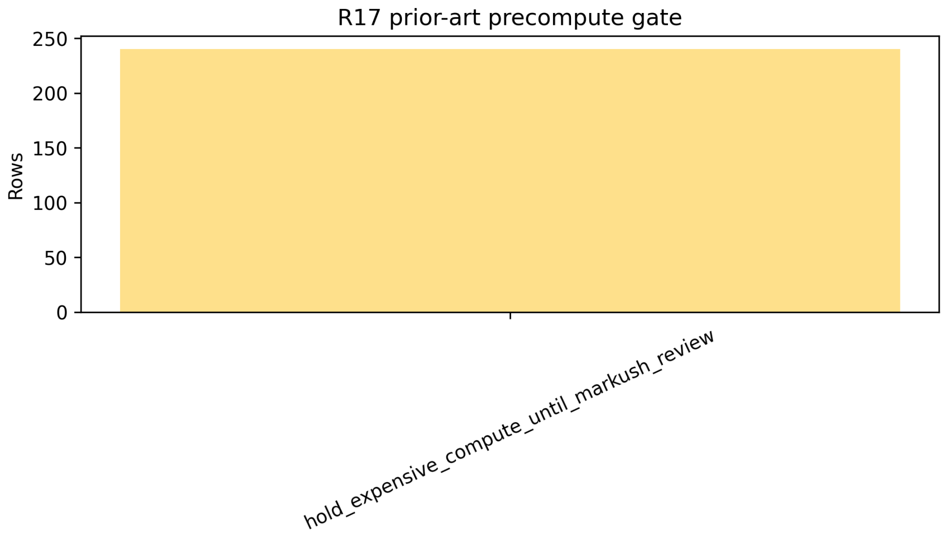

Figure 4. R17 prior-art precompute gate distribution.

4. Development Interpretation

R17 is best interpreted as a constrained hit-to-lead expansion paper. Its strongest contribution is a reproducible design queue that combines target cofolding, photosafety labeling, staged MD robustness, and prior-art discipline. R16 remains the cleaner immediate topical lead manuscript; R17 supplies the broader analog design space.

5. Limitations

All findings are in silico only. Boltz-2 cofolding is not biochemical binding evidence. MD RMSD stability is not potency, residence time, target engagement, or skin exposure. Photosafety, sensitization, hERG, AMES, DILI, irritation, IVRT/IVPT, PBPK, formulation compatibility, and target-engagement assays remain required. Markush/FTO review is mandatory before synthesis, purchasing, commercial novelty claims, or additional expensive 100-200 ns/RBFE/ABFE expansion.

6. Conclusion

The R17 atlas is complete for manuscript-level reporting as a disciplined constrained-design workflow: the 240-row cofold atlas and all top/next/expanded 10/30/60 ns MD panels are finished, including the final expanded 60 ns 3/3 stable gate. The next step is not another automatic long-MD expansion, but cross-model/decoy or PLIF checks, Markush/FTO review, and wet-lab/formulation packages.